Genetic Modification

Exam code: 0610 & 0970

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Genetic Modification: Definition

Genetic modification

- Genetic modification is **changing the genetic material of an organism by removing, changing or inserting individual genes from another organism**
- The organism receiving the genetic material is said to be **'genetically modified'**, or is described as a **'transgenic organism'**
- The DNA of the organism that now contains DNA from another organism as well is known as **'recombinant DNA'**

Help me understand

Genetic Modification: Examples

- There are many examples of genetically modified organisms, including:
- The gene for **human insulin** has been inserted into **bacteria** which then produce human insulin which can be collected and purified for medical use for diabetics
- **Crop plants**, such as wheat and maize, have been genetically modified to contain a gene from a bacterium

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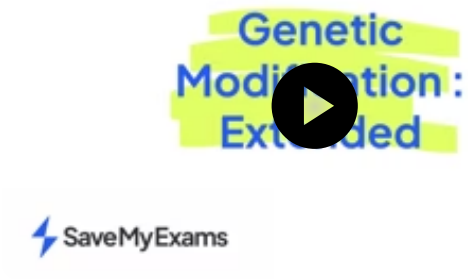
them **resistant to certain herbicides** (chemicals that kill plants), meaning that when the herbicide is sprayed on the crop it only kills weeds and does not affect the crop plant
- Some crops have been genetically modified to **produce additional vitamins**, eg **'golden rice'** contains genes from another plant and a bacterium which make the rice grains produce a chemical that is turned into **vitamin A** in the human body, which could help prevent deficiency diseases in certain areas of the world



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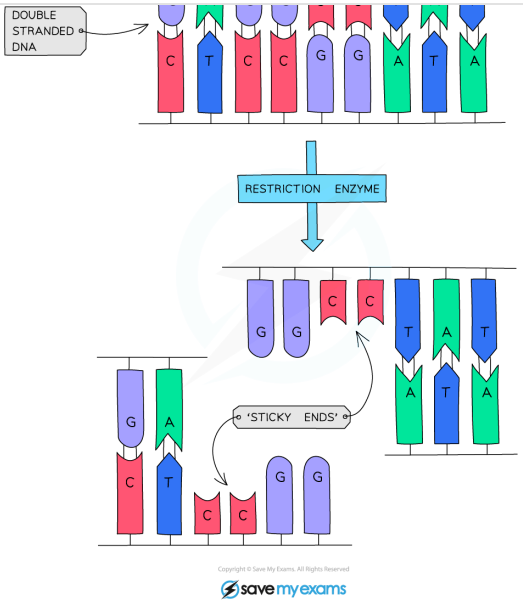
Genetic Modification: Examples : Extended

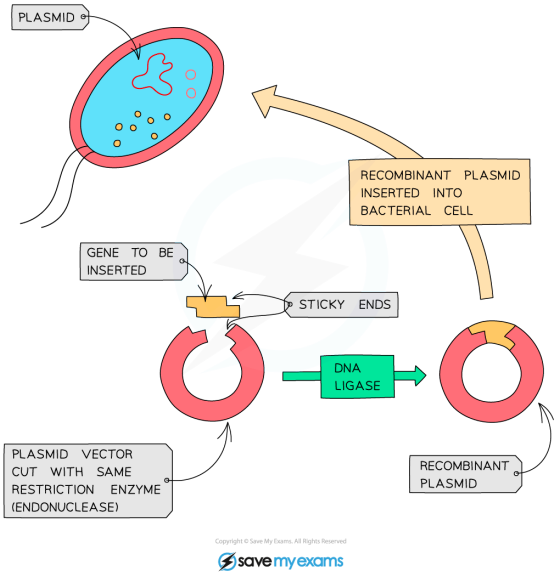
Genetic modification using bacterial production of a human protein

- The gene that is to be inserted is located in the original organism (for example, this could be the gene for human insulin)
- **Restriction enzymes** are used to isolate the required gene, leaving it with **'sticky ends'** (a short section of unpaired bases)
- A bacterial plasmid is **cut by the same restriction enzyme leaving it with corresponding sticky ends** (plasmids are **circles of DNA** found inside bacterial cells)



Restriction enzymes cut DNA strands at specific sequences to form 'sticky ends'

- The plasmid and the isolated gene are joined together by **DNA ligase enzyme**
- If two pieces of DNA have **matching sticky ends** (because they have been **cut by the same restriction enzyme**), DNA ligase will link them to form a single, unbroken molecule of DNA



DNA ligase is used to join two separate pieces of DNA together

- The genetically engineered plasmid is **inserted into a bacterial cell**
- When the bacteria reproduce the **plasmids are copied as well** and so a recombinant plasmid can **quickly be spread**

fermenter to reproduce quickly in controlled conditions and make **large quantities** of the human protein

- Bacteria are extremely useful for genetic engineering purposes because:
  - They contain the **same genetic code** as the organisms we are taking the genes from, meaning they can easily 'read' it and **produce the same proteins**
  - There are **no ethical concerns over their manipulation and growth** (unlike if animals were used, as they can feel pain and distress)
  - The **presence of plasmids** in bacteria, separate from the main bacterial chromosome, makes them **easy to remove and manipulate** to insert genes into them and then place back inside the bacterial cells

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